HIAST at IslamicEval 2026: Context-Aware Hallucination Detection and Evidence-Guided Verification for Arabic Islamic Question Answering

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Abstract

Large Language Models (LLMs) have significantly improved general-purpose question answering but remain prone to hallucinations in fact-sensitive domains such as religion. IslamicEval 2026 addresses these challenges by providing a fine-grained benchmark for evaluating hallucination and relevance in Arabic Islamic citations. This paper presents our approaches to three tasks: Citation Span Detection (Subtask 1), Hallucination Identification (Subtask 2), and Answer Relevance (Subtask 4). Across these tasks, we adopt a task-specific context modeling strategy in which contextual evidence is tailored to the semantic requirements of each citation understanding task. For **Citation Span Detection**, we formulate the problem as BIO-based token classification and fine-tune AraBERT using overlapping sliding-window encoding to process long responses effectively. For **Hallucination Identification**, we introduce a segment-level verification framework that augments each citation with adaptively extracted contextual evidence and citation-type information. For **Answer Relevance**, we combine hybrid lexical-semantic retrieval over authenticated Qur'anic and Hadith corpora with Transformer-based classification, enriching retrieved evidence with citation-specific context such as neighboring Ayat and explicit Surah–verse references. On the official test set, the proposed solutions achieved an F1 of 0.9725, an accuracy of 86.62%, and a macro-F1 of 0.7721 for Subtasks 1, 2, and 4, respectively, demonstrating the effectiveness of the proposed approaches.

1 Introduction

Large Language Models (LLMs) have made significant advances in question answering but remain prone to hallucinations in fact-sensitive domains (El Ganadi et al., 2025). This challenge is particularly critical for religious texts, where both citation authenticity and relevance are essential (Hamed et al., 2025). Even minor alterations to Qur'anic verses or Hadiths, as well as authentic yet irrelevant citations, can lead to misleading responses (Fawzi et al., 2026). Consequently, reliable Islamic question answering requires systems that can accurately identify citation boundaries, verify citation correctness, and assess citation relevance (Mubarak et al., 2025).

The IslamicEval 2026 benchmark (Alharbi et al., 2026) is designed to evaluate hallucination and relevance in Arabic Islamic citations generated by LLMs. It consists of Arabic question–answer pairs containing Qur'anic verses, Hadith texts, isnads, and claimed source references. IslamicEval 2026 evaluates multiple subtasks: Citation Span Detection (Subtask 1), which aims to locate and classify citation spans in responses generated by LLMs; Hallucination Identification (Subtask 2), which focuses on verifying the correctness of the citation; Hallucination Correction (Subtask 3), which focuses on providing the canonical form for citation; and Answer Relevance (Subtask 4), which evaluates whether the evidence cited adequately supports the answer to the given question.

In this paper, we present our approach [1] to Subtasks 1, 2, and 4 of IslamicEval 2026. Our approach is based on a task-specific context modeling strategy that adapts contextual evidence to the semantic requirements of each task. For Citation Span Detection, we formulate the task as BIO-based token classification and fine-tune AraBERT using an overlapping sliding-window mechanism that preserves contextual continuity across long responses. For Hallucination Identification, we propose a citation-centered verification framework that enriches each target citation with adaptive contextual evidence and citation-type information. For Answer Relevance, we introduce an evidence-guided framework

[1] Code is available at https://github.com/rsonbol/Arabic-NLP-for-Islamic-Content.

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that combines hybrid lexical–semantic retrieval with citation-aware evidence enrichment from authenticated Qur'anic and Hadith corpora.

Our main contributions are threefold: (1) a task-specific context modeling strategy for citation detection, verification, and relevance assessment; (2) a citation-aware hallucination verification framework combining adaptive context expansion and citation-type encoding; and (3) an evidence-guided relevance framework that integrates hybrid retrieval with enriched Islamic evidence for citation relevance assessment.

2 Related Work

The increasing use of large language models (LLMs) in religious applications has raised concerns regarding hallucinations and the reliability of generated Islamic content. To address these challenges, recent studies have focused on identifying, verifying, and correcting religious references in LLM-generated responses (Mubarak et al., 2025; Alharbi et al., 2026). Isnad AI (Elden, 2025) employed a fine-tuned AraBERTv2 model supported by a rule-based synthetic data generation pipeline for religious reference localization. TCE (ElKoumy et al., 2025) investigated retrieval-augmented prompting for citation verification, while BurhanAI (Al Adel et al., 2025) combined synthetic augmentation, diacritic variation, and morphological normalization with a fine-tuned language model to improve the detection of Qur'anic and Hadith references. HUMAIN (Omayrah et al., 2025) formulated span identification as a sequence-to-sequence annotation task and performed verification using a combination of retrieval-based similarity and exact matching.

While prior work highlights the importance of contextual grounding, different citation understanding tasks require different forms of context. We therefore adopt a task-specific context modeling strategy.

3 System Description

In this section, we present the methodologies developed for each of the three benchmark subtasks.

3.1 Citation Span Detection

We formulate Citation Span Detection as a token-level sequence labeling task using AraBERT[2] (Antoun et al., 2020) and the BIO tagging scheme. Each generated response is converted into token-level labels drawn from a label set of nine classes: `O`, `B-AYAH`, `I-AYAH`, `B-MATN`, `I-MATN`, `B-ISNAD`, `I-ISNAD`, `B-CLAIMED_SOURCE`, and `I-CLAIMED_SOURCE`, corresponding to the beginning and continuation of four citation categories: Ayah, Matn, Isnad, and Claimed Source. This formulation enables the model to jointly identify citation boundaries and predict citation types.

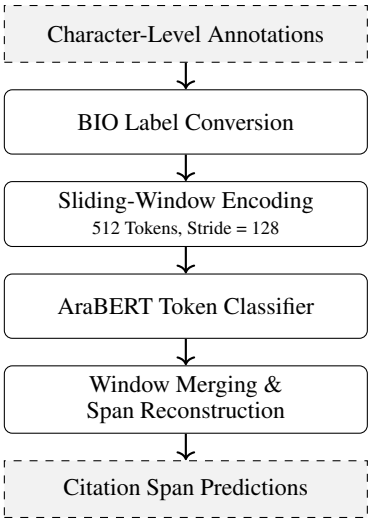


Figure 1: Citation span detection pipeline.

Documents longer than the model's input limit are processed using overlapping 512-token windows with a stride of 128. Character-level citation annotations are mapped to tokens via offset mappings and converted into BIO labels. Each window is treated as an independent training instance while preserving its association with the original document.

We fine-tune AraBERT as a token classification model using the generated BIO labels and optimize it with token-level cross-entropy loss. The overlap between adjacent windows helps preserve contextual information and reduces the likelihood of splitting citation spans across window boundaries.

During inference, the model predicts the BIO labels for each token, which are decoded into citation spans using the corresponding offset mappings. Predictions from overlapping windows are merged at the document level, duplicate spans are removed, and contiguous token sequences sharing the same label are reconstructed into complete citation spans. Finally, the extracted spans are converted into character-level offsets and formatted according to the official evaluation protocol.

[2] `aubmindlab/bert-base-arabertv02`

3.2 Hallucination Identification

In Subtask 2, citation boundaries are provided by the benchmark, shifting the task from citation localization to factual verification. Rather than classifying the entire generated response, we transform each annotated citation into an independent verification instance by explicitly marking its boundaries and citation type using dedicated special tokens, namely `<Ayah>`, `<Matn>`, `<Isnad>`, and `<ClaimedSource>`, together with their corresponding closing tags.

To provide sufficient evidence while avoiding unnecessarily long inputs, we construct a citation-centered context representation. The sentence containing the target citation is first identified, after which neighboring sentences are incrementally added until a budget of approximately 650 tokens is reached. Context expansion prioritizes sentences following the citation before considering preceding sentences, preserving the natural explanatory flow of the response.

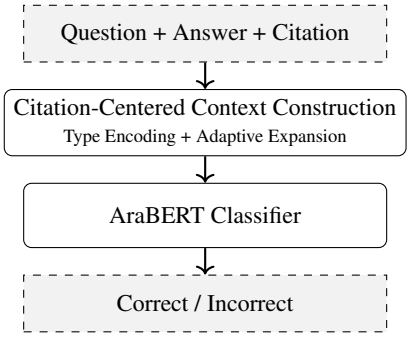


Figure 2: Hallucination identification pipeline.

The resulting input consists of the user question paired with the context-enhanced response containing the highlighted citation. We fine-tune an AraBERT sequence classification model to predict whether the marked citation is *Correct* or *Incorrect*. To avoid data leakage, we use `GroupShuffleSplit` at the response level, ensuring that all citations from a given response remain within the same split. The question and citation-centered context are encoded as paired inputs, with truncation applied only to the context when necessary. To mitigate label imbalance, class weights are incorporated into the cross-entropy loss function. Training further employs mixed-precision optimization and a linear learning-rate scheduler with warm-up, while the checkpoint achieving the highest validation Macro-F1 score is selected as the final model.

3.3 Answer Relevance

Answer Relevance aims to determine whether an authentic Islamic citation adequately addresses the user's question. To reduce reliance on the classifier's parametric knowledge, we adopt a retrieval-augmented, evidence-grounded framework that explicitly verifies each citation against supporting evidence retrieved from trusted Islamic sources.

An *Evidence Builder* first constructs a structured evidence representation for each claim. Given a question, generated answer, and target citation, the system extracts the most relevant supporting sentences from the generated answer using semantic similarity. It then retrieves corroborating evidence from both Qur'anic and Hadith corpora through a hybrid retrieval strategy that combines dense semantic retrieval using `multilingual-e5-large` embeddings (Wang et al., 2024) indexed with FAISS (Douze et al., 2025) and sparse lexical retrieval using BM25. The outputs of both retrieval components are integrated through Reciprocal Rank Fusion (RRF) to improve retrieval robustness. For Qur'anic evidence, the retrieved verse is further enriched with its surrounding context by incorporating neighboring Ayat within a fixed context window, while citation metadata such as Surah names and Ayah numbers are preserved whenever available.

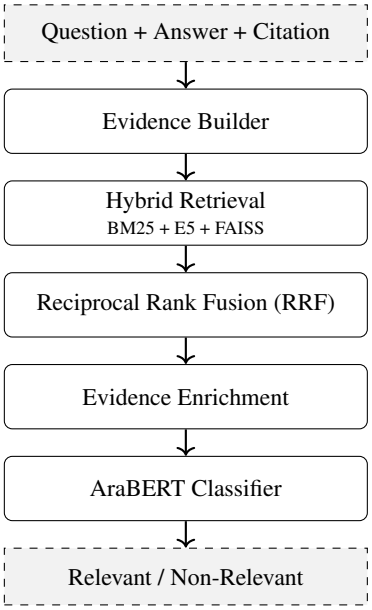


Figure 3: Answer relevance pipeline.

The resulting evidence is organized into a structured input containing four task-specific markers: `[QUESTION]`, `[ANSWER]`, `[CLAIM]`, and `[EVIDENCE]`. The `[EVIDENCE]` section includes rel-

evant supporting sentences from the generated answer together with the retrieved Qur'anic and Hadith evidence. This evidence-aware representation is then processed by a fine-tuned `AraBERTv02` classifier, which jointly models the question, answer, citation, and authenticated evidence to predict whether the citation is *Relevant* or *Non-Relevant*.

4 Evaluation and Results Analysis

All experiments were conducted using the official IslamicEval 2026 datasets and evaluation protocol. To ensure consistency across subtasks, all systems were built on AraBERT v0.2 and fine-tuned using a maximum sequence length of 512 tokens, batch size 8, AdamW optimization, linear learning-rate scheduling, and early stopping. Class-weighted loss was used for Subtask 2 to mitigate class imbalance.

4.1 Citation Span Detection

Our system achieved an **F1-score of 97.25%**, demonstrating that citation boundaries can be effectively modeled through token-level sequence labeling. The strong performance indicates that citation spans in the benchmark exhibit relatively consistent lexical and structural patterns, enabling AraBERT to accurately identify both span boundaries and citation types, even in long responses. By formulating citation span detection as a BIO-based token classification task, the model can predict citation boundaries and categories within a unified framework.

4.2 Hallucination Identification

Table 1 presents the official evaluation results of our system for Subtask 2. The submitted model achieved an **overall accuracy of 86.62%**. Although performance varies across citation categories, the observed differences remain relatively small (3.44 percentage points). Ayah and Matn citations achieve the highest accuracies, likely because they follow more regular linguistic and stylistic patterns. In contrast, Isnad citations comprise the most challenging category, as they often contain long chains of narrators and more diverse lexical structures, making factual verification more difficult.

4.3 Answer Relevance

The system was evaluated using the official Task 4 metric, Macro-F1. Our approach achieved a **Macro-F1 score of 0.7721** on the official test set.

Citation Type	Accuracy (%)
Ayah	87.74
Matn	87.55
Claimed Source	86.89
Isnad	84.30
Overall	**86.62**

Table 1: Official test results for Subtask 2 (Hallucination Identification).

In particular, 165 out of 271 questions (60.9%) achieved a perfect F1 score of 1.0, while 192 questions (70.8%) obtained an F1 score of at least 0.8. These results show that the proposed evidence-grounded framework is effective for relevance assessment, suggesting that hybrid retrieval and citation-aware context provide complementary signals for determining whether a citation adequately supports a generated answer. However, errors were frequently observed when the retrieved evidence was relevant to the question but provided only indirect support for the generated answer. Such cases highlight the challenges of distinguishing semantic relatedness from true evidential relevance.

5 Conclusion

In this paper, we presented our participation in IslamicEval 2026 for Citation Span Detection, Hallucination Identification, and Answer Relevance. Our approaches leverage task-specific context modeling for citation detection, verification, and relevance assessment. The proposed approaches achieved strong performance on the official test set, obtaining an F1 score of 0.9725 for Citation Span Detection, an accuracy of 86.62% for Hallucination Identification, and a Macro-F1 score of 0.7721 for Answer Relevance. These results demonstrate the effectiveness of adapting contextual representations to the semantic requirements of different citation understanding tasks.

Limitations

The proposed approaches were developed and evaluated within the IslamicEval benchmark and may not generalize directly to other domains, languages, or citation styles. Furthermore, while our results demonstrate the effectiveness of the overall framework, a more comprehensive analysis of the contribution of individual components, along with extensive ablation studies, remains necessary. We leave these investigations for future work.

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